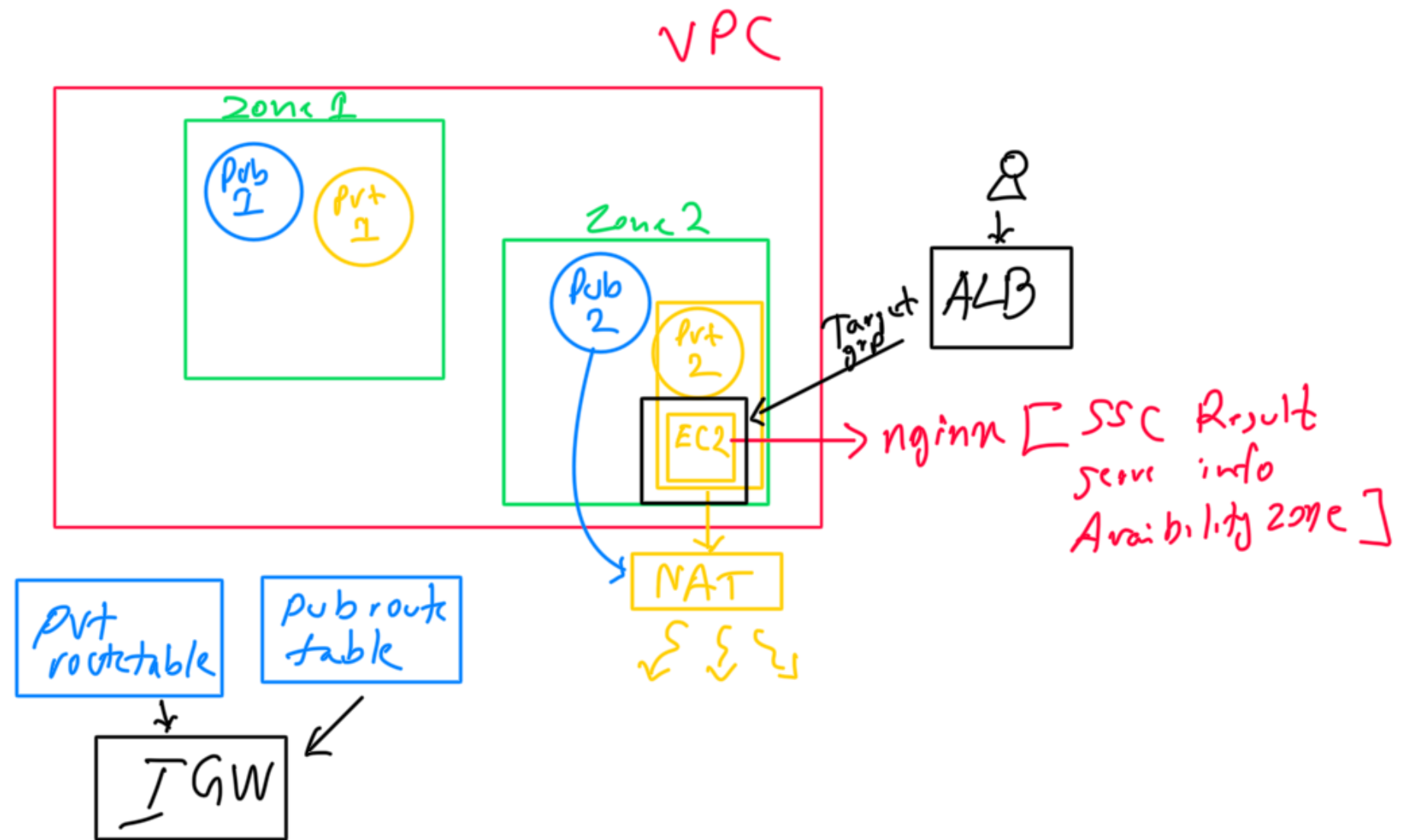




# Scalable SSC Result Publishing Infrastructure

AWS Practical Assignment — Submission Report

Scope: VPC → Private EC2 (Nginx) → Application Load Balancer → Access via ALB DNS  
(Auto Scaling configuration excluded from this submission, per scope agreed for this phase)

Region: Asia Pacific (Mumbai) — ap-south-1

# Short Report

## 1. VPC / Network Architecture

A custom VPC named **Geralt VPC-vpc** was created with CIDR block 10.0.0.0/16 in the ap-south-1 (Mumbai) region. The VPC contains four subnets spread across two Availability Zones for high availability:

- Geralt VPC-subnet-public1-ap-south-1a and Geralt VPC-subnet-public2-ap-south-1b — public subnets that host the Application Load Balancer and the bastion host.
- Geralt VPC-subnet-private1-ap-south-1a and Geralt VPC-subnet-private2-ap-south-1b — private subnets that host the application (Nginx) EC2 instances, with no public IP addresses assigned.

An Internet Gateway (Geralt VPC-igw) is attached to the VPC to allow public-subnet resources to communicate directly with the internet. Two route tables are used to separate public and private traffic paths: the public route table sends 0.0.0.0/0 traffic to the Internet Gateway, while the private route table sends 0.0.0.0/0 traffic to the NAT Gateway. Both route tables keep the local 10.0.0.0/16 route for intra-VPC traffic.

## 2. ALB and EC2 Traffic Flow

Traffic flows in a strict Internet → ALB → EC2 pattern:

- The Application Load Balancer (Geralt) is internet-facing and deployed across the two public subnets, so it is reachable from the internet on port 80.
- The ALB listener (HTTP:80) forwards all matching requests to the target group (Geralt), which is configured with target type "Instance" and protocol HTTP on port 80.
- The EC2 instance running Nginx sits in a private subnet and has only a private IP address (10.0.141.150) — it has no route to or from the internet directly.
- The EC2 instance's security group only needs to accept HTTP traffic on port 80 for the ALB to reach it; the instance itself is never exposed publicly because it has no public IP and no direct internet route (its subnet routes 0.0.0.0/0 to the NAT Gateway, not the Internet Gateway).
- SSH management access to the private EC2 instance is done indirectly through a bastion host (Geralt Bastion) placed in the public subnet — administrators SSH into the bastion first, then hop to the private instance using its private IP, keeping the private EC2 fully unreachable from the public internet.

This satisfies the security requirement that EC2 instances must not be directly accessible from the internet, while still allowing end users to reach the application through the ALB's public DNS name.

## 3. NAT Gateway Purpose

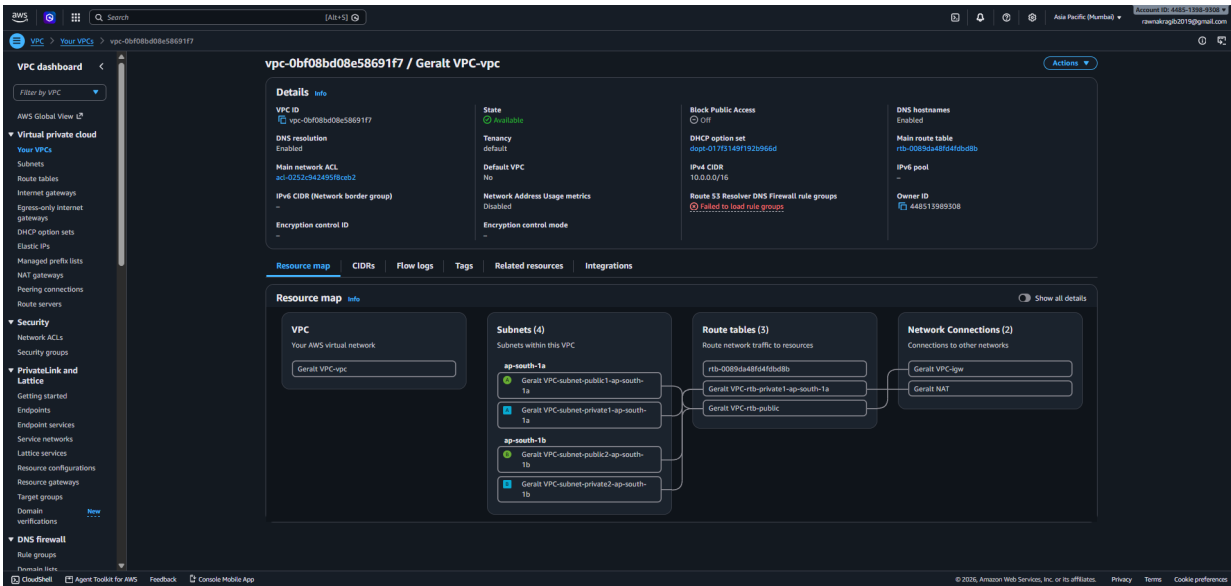
The NAT Gateway (Geralt NAT) is deployed in a public subnet and referenced as the default route (0.0.0.0/0) in the private route table. It allows instances in the private subnets to initiate outbound connections to the internet — for example, to run "sudo apt update" and install/configure Nginx during setup — without exposing those instances to unsolicited inbound internet traffic. Unlike an Internet Gateway, a NAT Gateway only allows outbound-initiated traffic, which keeps the private subnet secure while still enabling necessary software installation and patching.

## 4. Handling the SSC Result Traffic Spike

For this submission, the infrastructure demonstrates the foundational layer that any scaling strategy sits on top of: a load-balanced, privately-hosted web tier. The ALB distributes incoming requests across registered EC2 targets and performs health checks on the target group, so unhealthy instances can automatically be taken out of rotation without affecting availability. This design is ready to be extended with an Auto Scaling Group (minimum 2, desired 2, maximum 10+) attached to the same target group, so that during the 10:00 AM result-publishing spike, additional EC2 instances can be launched automatically from the Launch Template across both Availability Zones and registered behind the same ALB — and scaled back in once traffic subsides. Auto Scaling configuration itself is out of scope for this submission.

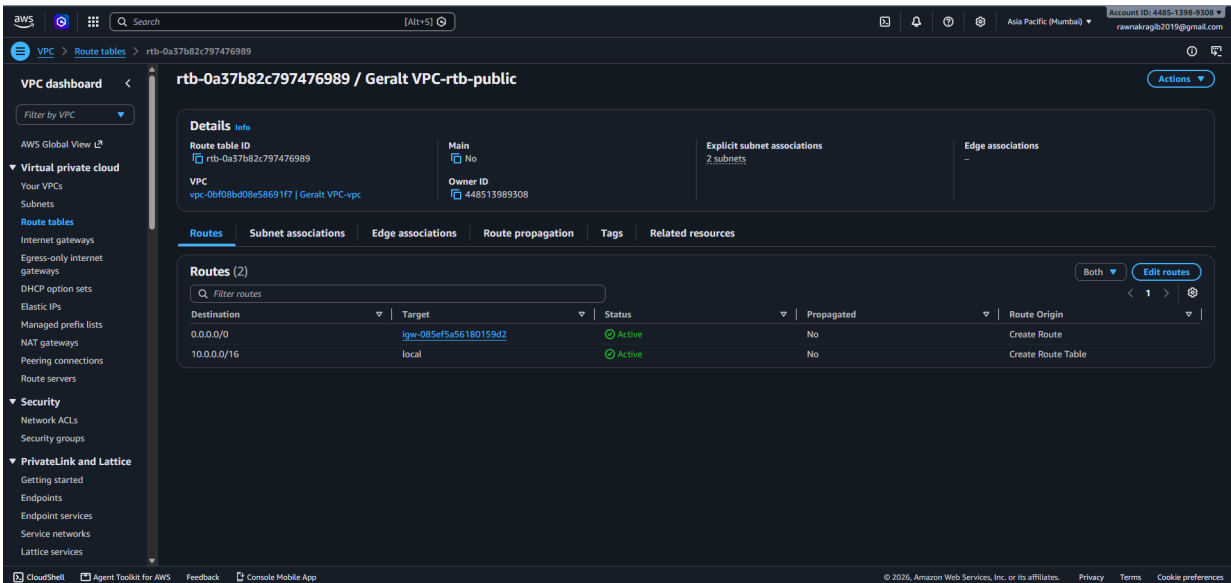
# Screenshots

## 1. VPC — Resource Map



Geralt VPC-vpc (10.0.0.0/16) showing 4 subnets across 2 Availability Zones, 3 route tables, and network connections (Internet Gateway and NAT Gateway).

## 2. Public Route Table



Geralt VPC-rtb-public: routes 0.0.0.0/0 to the Internet Gateway (igw-...) and 10.0.0.0/16 locally.

### 3. Private Route Table

**rtb-0112b9caa8ac961cb / Geralt VPC-rtb-private1-ap-south-1a**

**Details**

- Route table ID: rtb-0112b9caa8ac961cb
- VPC: vpc-0b08bd08e58691f7 / Geralt VPC-vpc
- Main: No
- Owner ID: 448513989308
- Explicit subnet associations: subnet-0e24f9ee44c21262f / Geralt VPC-subnet-private1-ap-south-1a
- Edge associations: -

**Routes (2)**

| Destination | Target                | Status | Propagated | Route Origin       |
|-------------|-----------------------|--------|------------|--------------------|
| 0.0.0.0/0   | nat-009d11c29582a6665 | Active | No         | Create Route       |
| 10.0.0.0/16 | local                 | Active | No         | Create Route Table |

Geralt VPC-rtb-private1-ap-south-1a: routes 0.0.0.0/0 to the NAT Gateway (nat-...) and 10.0.0.0/16 locally, keeping the private subnet's outbound traffic NAT'd.

### 4. NAT Gateway

**nat-009d11c29582a6665 / Geralt NAT**

**Details**

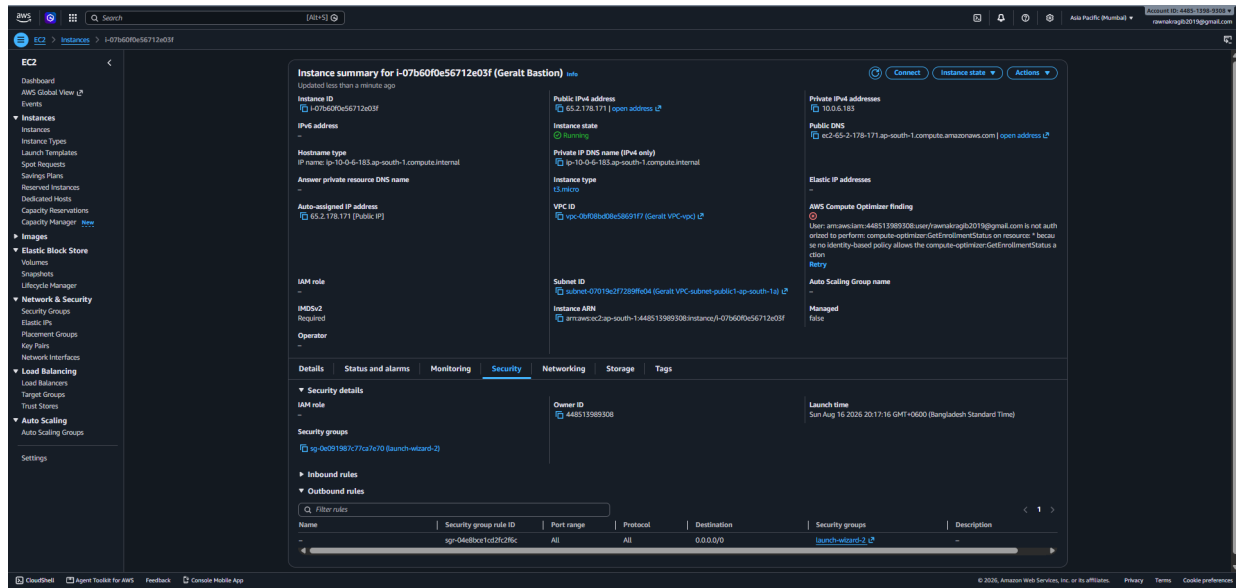
- NAT gateway ID: nat-009d11c29582a6665
- NAT gateway ARN: arn:aws:ec2:ap-south-1:448513989308:natgateway/nat-009d11c29582a6665
- VPC: vpc-0b08bd08e58691f7 / Geralt VPC-vpc
- Connectivity type: Public
- Primary public IPv4 address: -
- Subnet: subnet-07019e2f7289ffe04 / Geralt VPC-subnet-public1-ap-south-1a
- State: Pending
- Primary private IPv4 address: 10.0.9.168
- Created: Sunday, August 16, 2026 at 19:55:06 GMT+6
- State message: -
- Primary network interface ID: eni-09f85c786e1644d20
- Deleted: -

**Secondary IPv4 addresses**

Secondary IPv4 addresses are not available for this nat gateway.

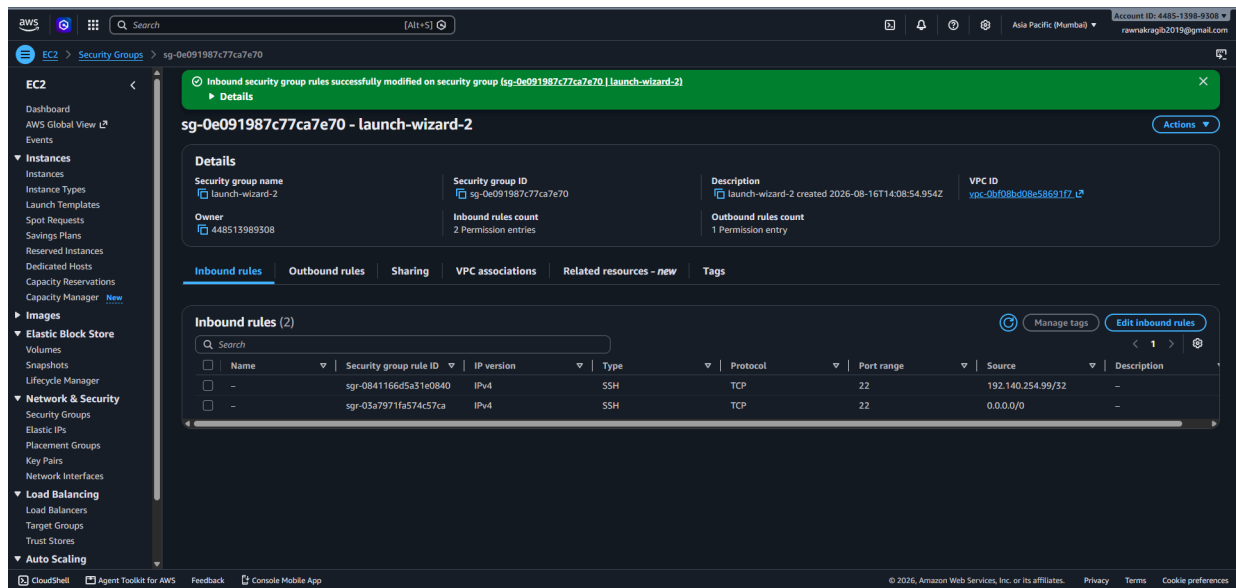
Geralt NAT gateway deployed in the public subnet (ap-south-1a), providing outbound internet access for the private subnets.

## 5. Bastion Host EC2 Instance



Geralt Bastion instance in the public subnet with a public IP (65.2.178.171), used solely as a secure jump host for SSH management access.

## 6. Bastion Host — Security Group Inbound Rules



SSH (port 22) restricted to specific known IP addresses, not open to the world.

## 7. SSH Login to Bastion Host

```
ubuntu@ip-10-0-6-183: ~$ ssh -i /home/ubuntu/.ssh/geralt.pem ubuntu@65.2.178.171
Warning: Permanently added '65.2.178.171' (ED25519) to the list of known hosts
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Aug 16 14:21:19 UTC 2026

System load:  0.0               Temperature:  -273.1 C
Usage of /:   30.5% of 6.61GB   Processes:    119
Memory usage: 25%              Users logged in: 0
Swap usage:   0%               IPv4 address for ens5: 10.0.6.183

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

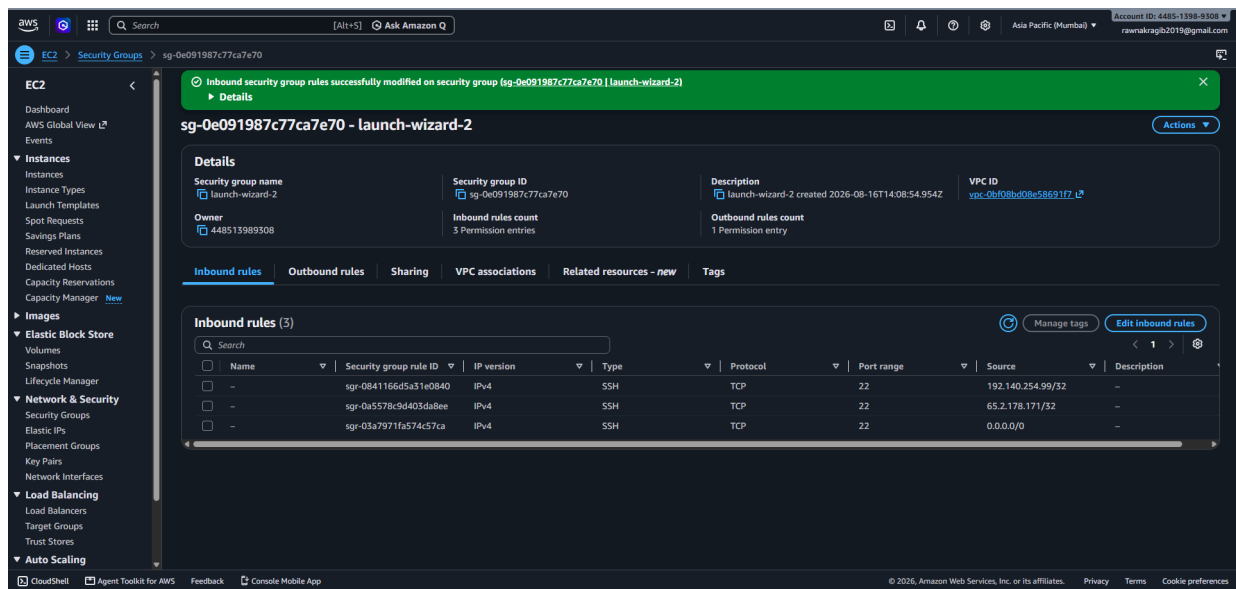
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-6-183:~$ |
```

Terminal session showing a successful SSH connection to the bastion host using its public IP.

## 8. Adding Bastion IP to Private EC2 Security Group



The private EC2's security group inbound rules, updated to allow SSH only from the bastion host's IP so the private instance can be reached solely via the bastion.



## 9. Connecting to Private EC2 via Bastion Host

```
ubuntu@ip-10-0-141-150:~$ ssh -i Geralt.pem ubuntu@13.201.116.35
AC
ubuntu@ip-10-0-6-183:~/.ssh$ ssh -i Geralt.pem ubuntu@10.0.141.150
The authenticity of host '10.0.141.150 (10.0.141.150)' can't be established.
ED25519 key fingerprint is: SHA256:w6BA/oiZ9pL013Xj2SpFz8bcD3ZU5ymbpUwDmv2npI
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.141.150' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun Aug 16 14:31:05 UTC 2026

System load:  0.0          Temperature:   -273.1 C
Usage of /:   30.5% of 6.61GB Processes:      115
Memory usage: 7%          Users logged in: 0
Swap usage:   0%          IPv4 address for ens5: 10.0.141.150

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

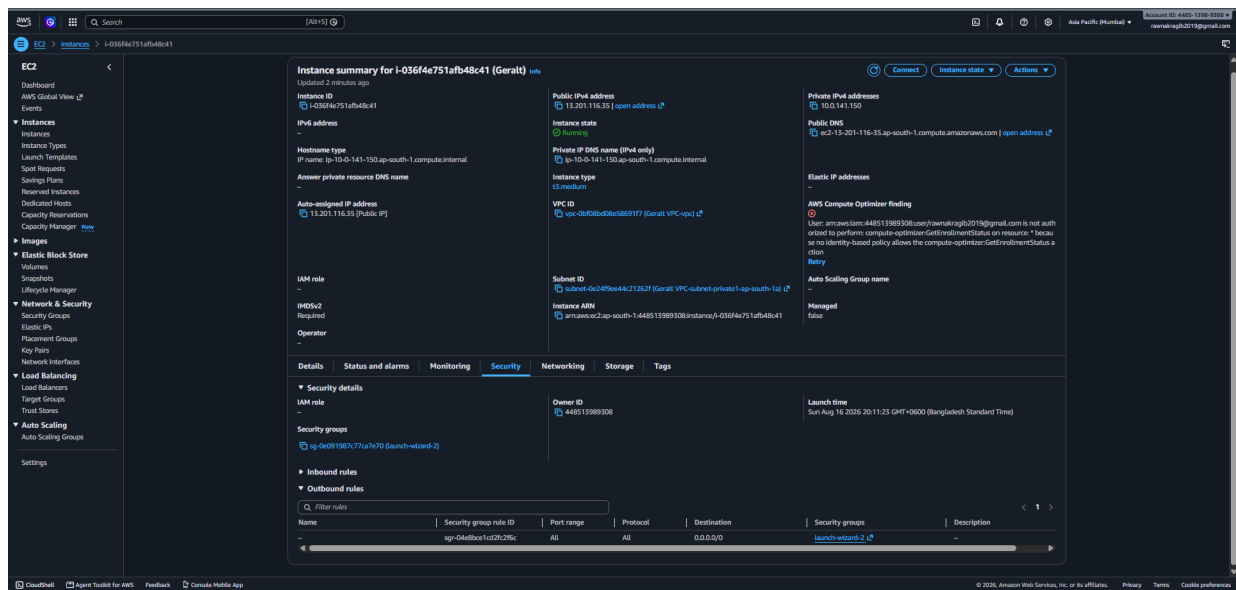
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-141-150:~$ |
```

From the bastion host, SSH-ing into the private EC2 instance (10.0.141.150) — demonstrating that the private instance is only reachable through the bastion, never directly from the internet.

## 10. Private EC2 Instance Details



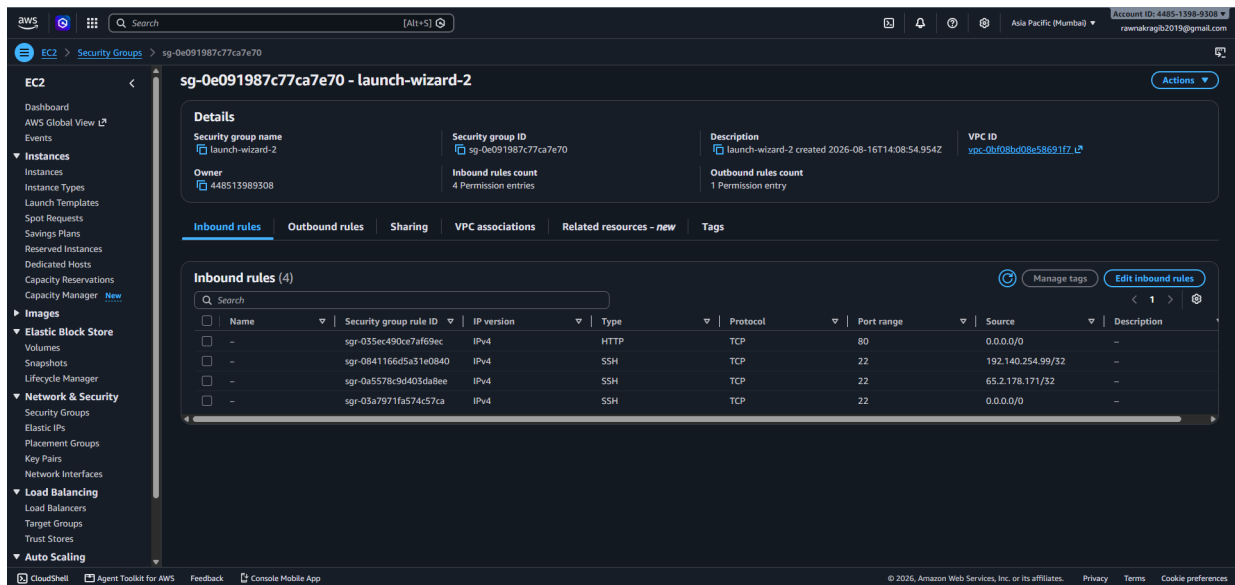
Geralt application instance running in a private subnet with only a private IP address (10.0.141.150) — no public IPv4 address is assigned.

## 11. Configuring the Nginx Web Page

```
ubuntu@ip-10-0-141-150:/var/www/html$ ls
index.nginx-debian.html
ubuntu@ip-10-0-141-150:/var/www/html$ sudo bash -c 'cat <<EOF > /var/www/html/index.nginx-debian.html
<DOCTYPE html>
<html>
<head>
<title>SSC Result 2026</title>
</head>
<body>
<h1>SSC Result 2026</h1>
<h2>Instance info:</h2>
<p>ID: i-036f4e751afb48c41<br>
IP: 10.0.141.150</p>
<p>Availability zone is Mumbai AP South 1a</p>
</body>
</html>
EOF'
```

Custom index.nginx-debian.html written on the private instance, containing the SSC Result 2026 title, instance ID/IP, and Availability Zone.

## 12. Enabling Port 80 for ALB Traffic



Security group updated with an HTTP (port 80) inbound rule so the Application Load Balancer can forward web traffic to the private EC2 instance.

## 13. Target Group

The screenshot shows the AWS Management Console for the 'Geralt' Target Group. The left sidebar contains navigation links for EC2, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'Details' tab for the 'Geralt' target group. The details include the ARN, Target type (Instance), Protocol (HTTP), Port (80), Protocol version (HTTP1), VPC (vpc-0bf08bd08e58691f7), and IP address type (IPv4). A summary row shows 1 Total target, 0 Healthy, 0 Unhealthy, 1 Unused, 0 Initial, and 0 Draining. Below this is a 'Distribution of targets by Availability Zone (AZ)' section. The 'Targets' tab shows a table with one registered target: 'i-036f4e751afb48c41' (Geralt) with port 80, in the 'ap-south-1a' zone, with a health status of 'Unused'.

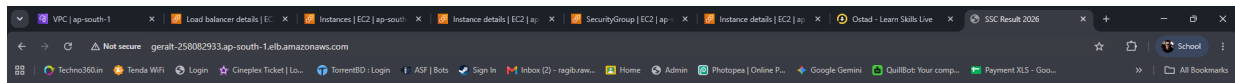
Geralt target group (HTTP:80, instance target type) with the private EC2 instance registered as a target.

## 14. Application Load Balancer

The screenshot shows the AWS Management Console for the 'Geralt' Application Load Balancer. The left sidebar contains navigation links for EC2, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'Details' tab for the 'Geralt' load balancer. The details include the Load balancer type (Application), Scheme (Internet-facing), Status (Active), Hosted zone (ZP978AFLXTHZK), VPC (vpc-0bf08bd08e58691f7), Availability Zones (subnet-07019e277289ff404, subnet-0ca4c78ffbd), Load balancer IP address type (IPv4), Date created (August 16, 2026, 21:02 UTC+06:00), and Load balancer ARN (arn:aws:elasticloadbalancing:ap-south-1:448513989308:loadbalancer/app/Geralt/52c196c8c8e49879). The 'Listeners and rules' tab shows a table with one listener: 'HTTP:80' with a default action of 'Forward to target group Geralt (2 of 1 (100%))' and a target group stickiness of 'Off'.

Geralt Application Load Balancer — internet-facing, deployed across two public subnets/Availability Zones, with an HTTP:80 listener forwarding to the Geralt target group.

## 15. Application Accessed via ALB DNS



### SSC Result 2026

#### Instance info:

ID: i-0366f4e751af8c41  
IP: 10.0.141.150  
Availability zone is Mumbai AP South 1a



The SSC Result 2026 page, served by Nginx on the private EC2 instance, successfully loaded in the browser using the Load Balancer's public DNS name (geralt-258082933.ap-south-1.elb.amazonaws.com) — confirming the Internet → ALB → EC2 flow works end-to-end.